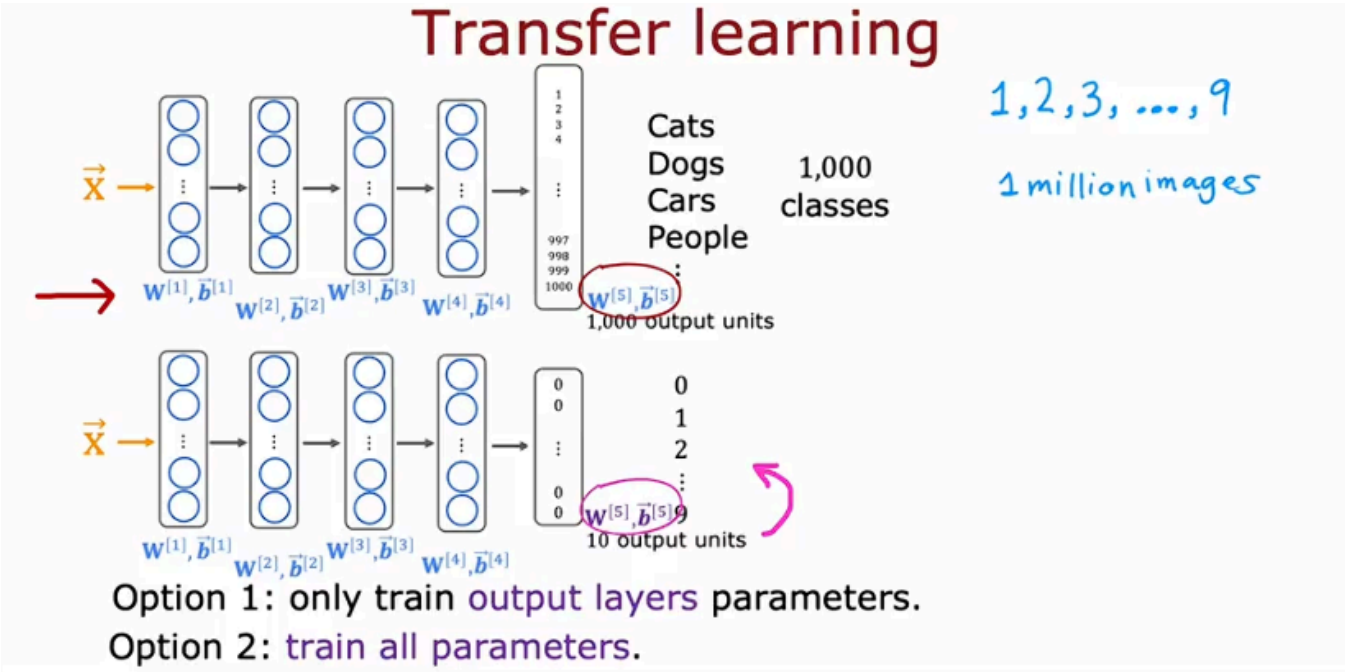
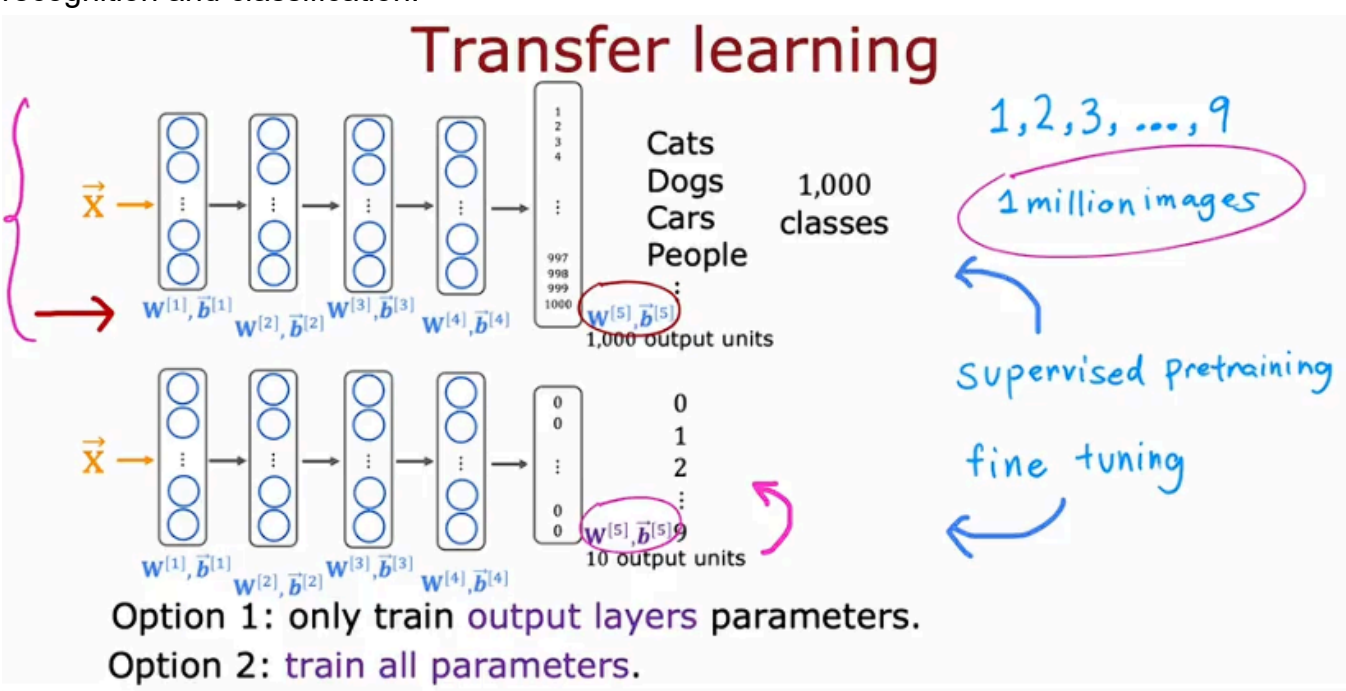
Day 23. Machine learning development process II

Machine learning development process

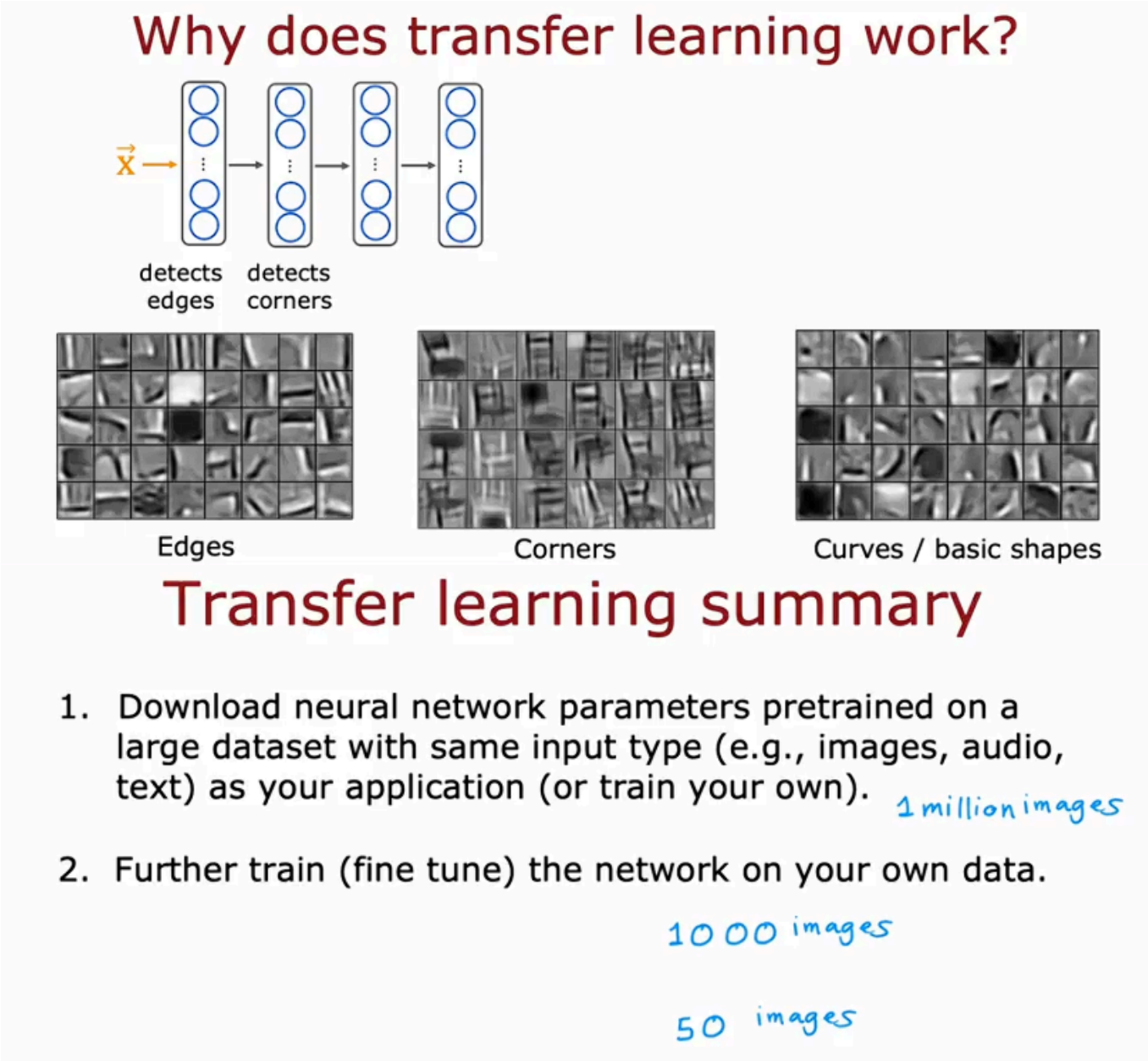
Transfer Learning



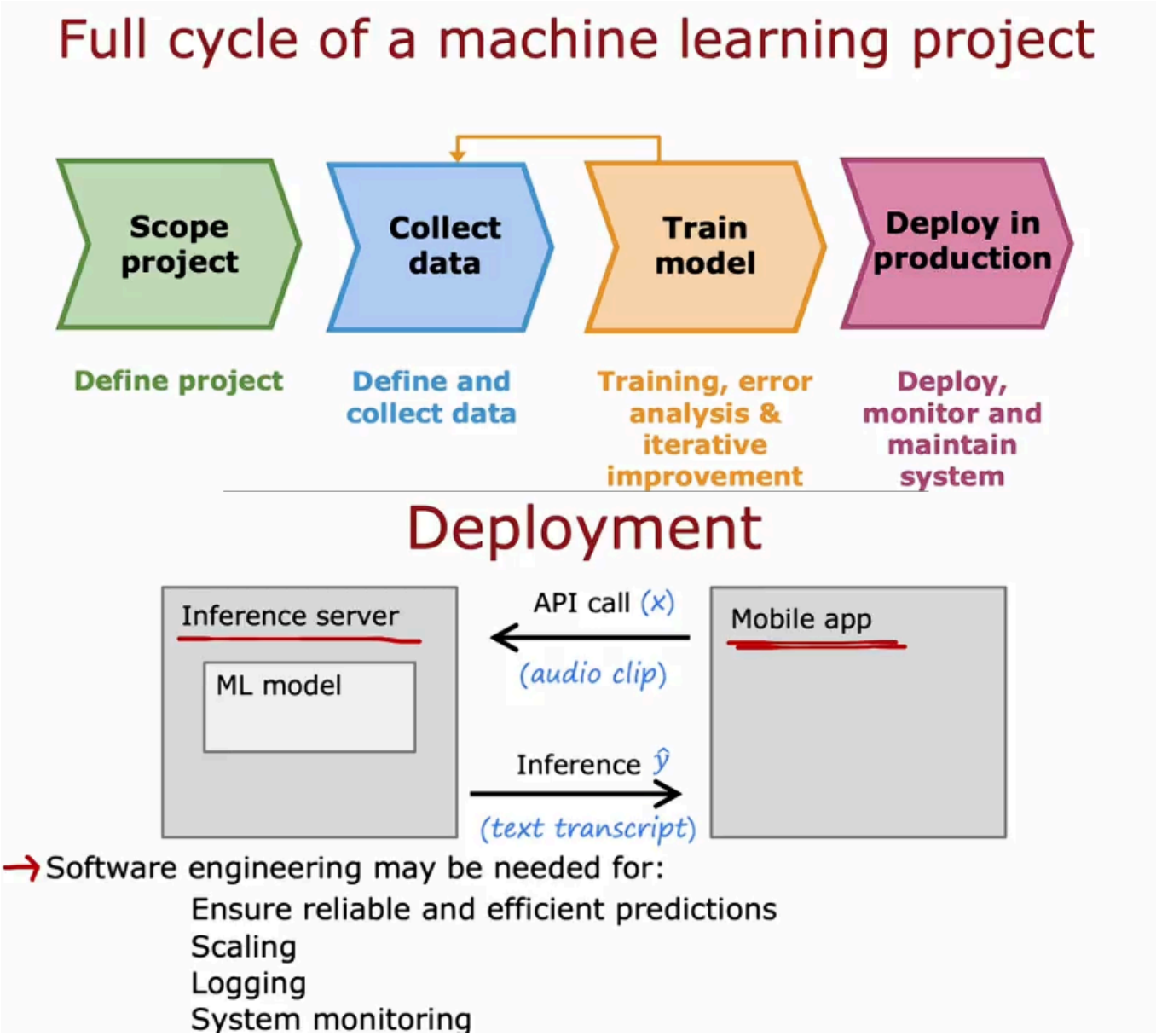
the intuition is that, by learning supersets, the model will get some basic parameters for recognition and classification.



we can download and fine tune a pretrained model trained by others to make better effects in your own field.



Full cycle of a machine learning project



Fairness, bias, and ethics

Bias

Bias

Hiring tool that discriminates against women.

Facial recognition system matching dark skinned individuals to criminal mugshots.

Biased bank loan approvals.

Toxic effect of reinforcing negative stereotypes.

Adverse use cases

Deepfakes

Adverse use cases

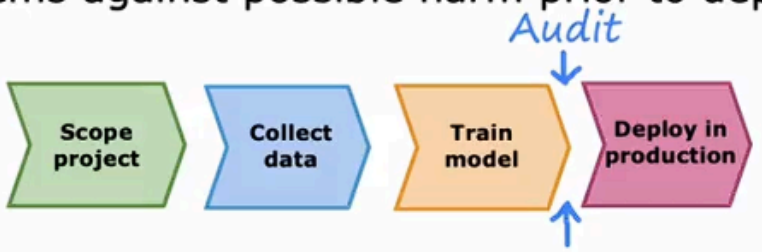
Deepfakes

Spreading toxic/incendiary speech through optimizing for engagement.

Generating fake content for commercial or political purposes.

Guidelines

Get a diverse team to brainstorm things that might go wrong, with emphasis on possible harm to vulnerable groups.

Carry out literature search on standards/guidelines for your industry.

Audit systems against possible harm prior to deployment.